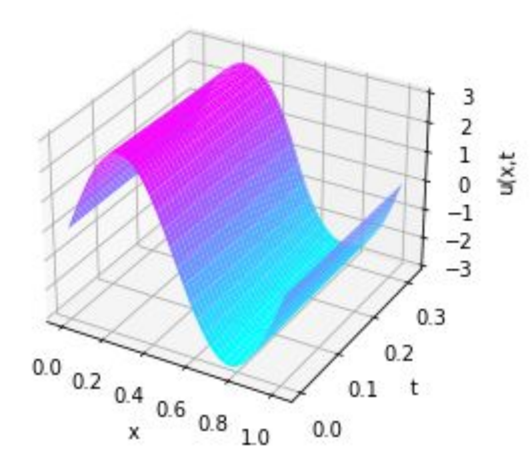
Université Paris Cité

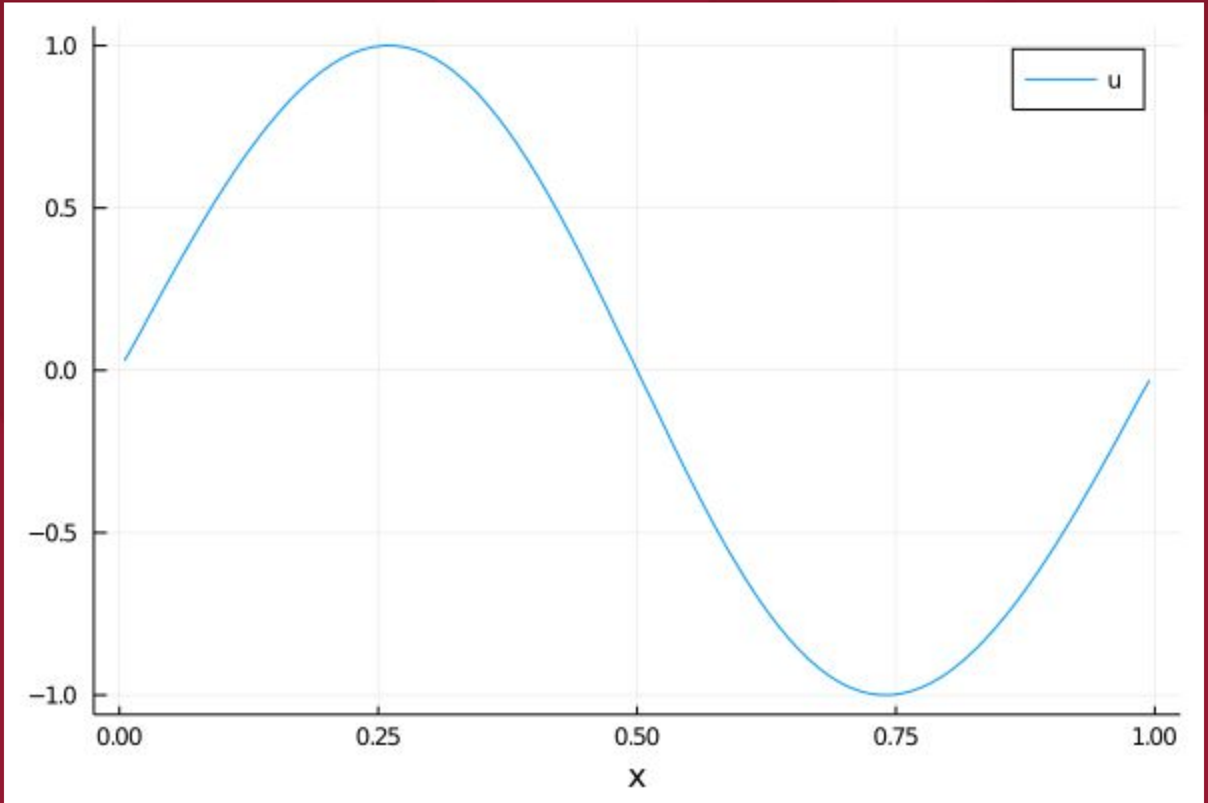
Résolution numérique de l'équation de Burgers

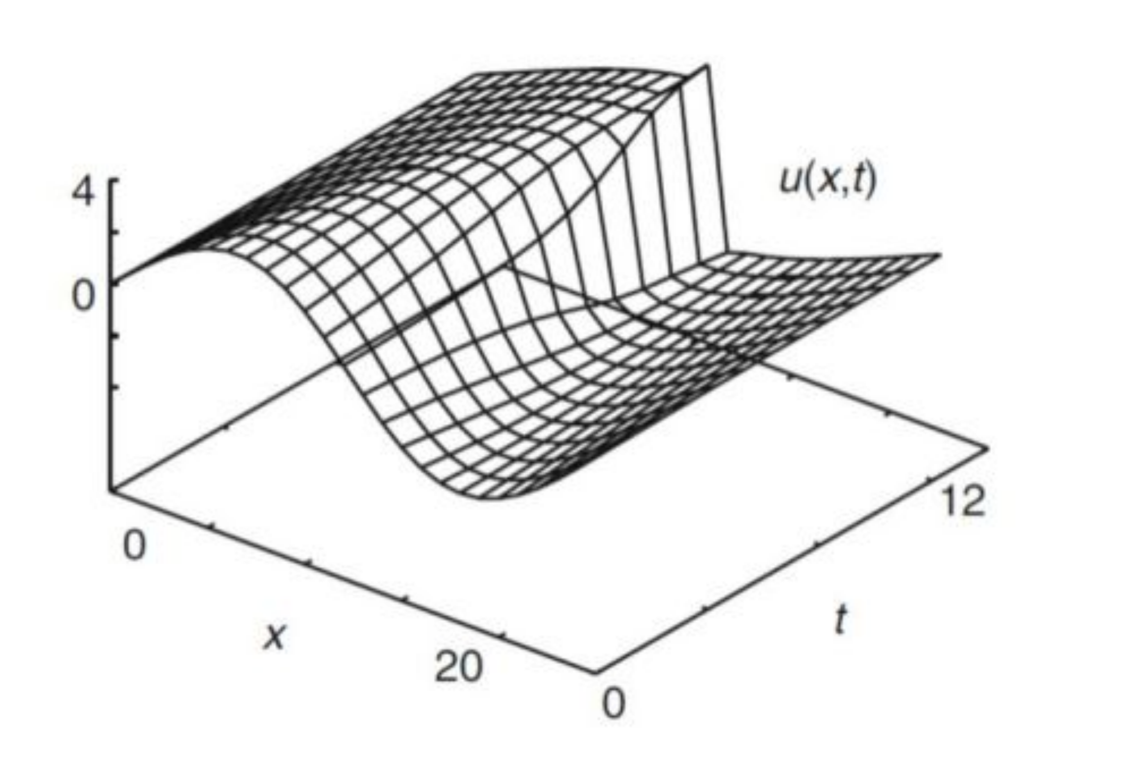
Physique numérique, L3 2023
– Marcos Morote Balboa

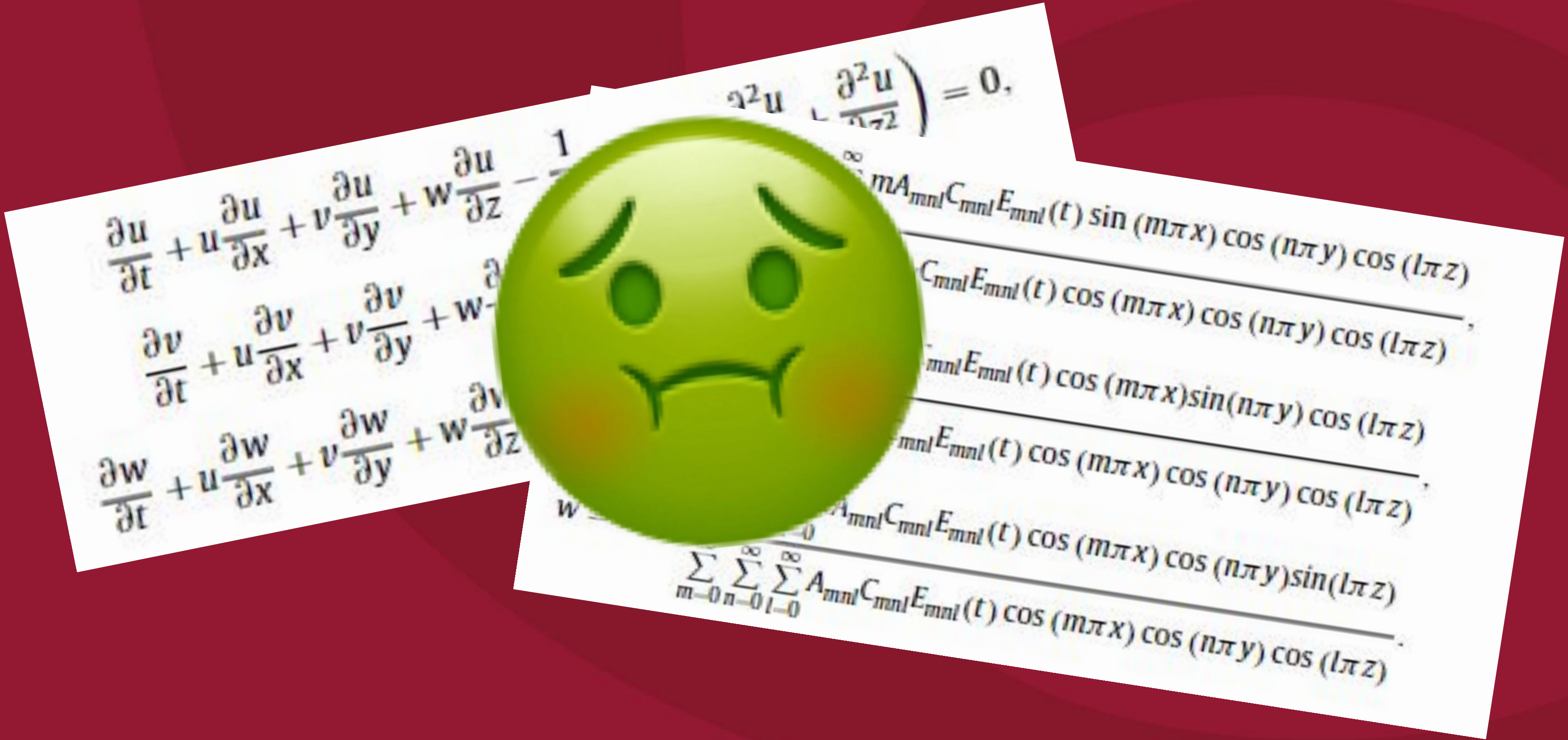


L'équation de Burgers

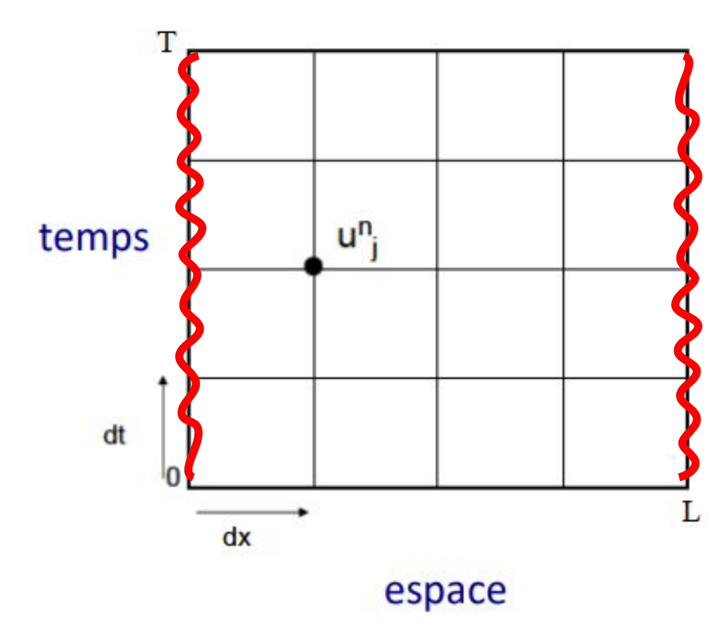
$$\boxed{\frac{\partial u}{\partial t} + \boxed{\varepsilon u}\frac{\partial u}{\partial x} = 0}$$


1.0
0.5
0.0
−0.5
−1.0
0.00
0.25
0.50
0.75
1.00
x
u


4
0
u(x,t)
0
x
20
0
t
12



Résolution



- $x \in [0, L]\ , t \in [0, T]$
- $\begin{cases} \Delta t = \dfrac{T}{N} \\ \Delta x = \dfrac{L}{N} \end{cases}$
- $u(0,t) = u(L,t) = 0 \ \ \forall t$
- $u(x) = Asin\left(\dfrac{2\pi x}{L}\right)$

Méthode nº1: *leapfrog*

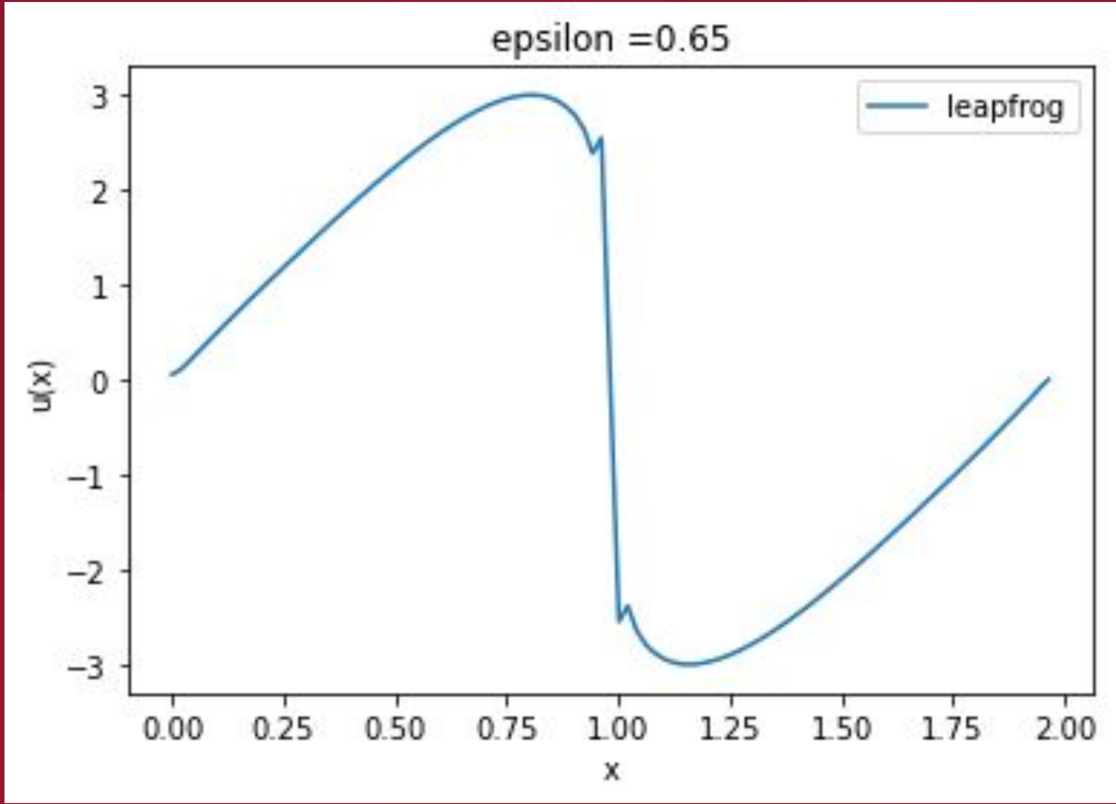
→ Différences finies centrées

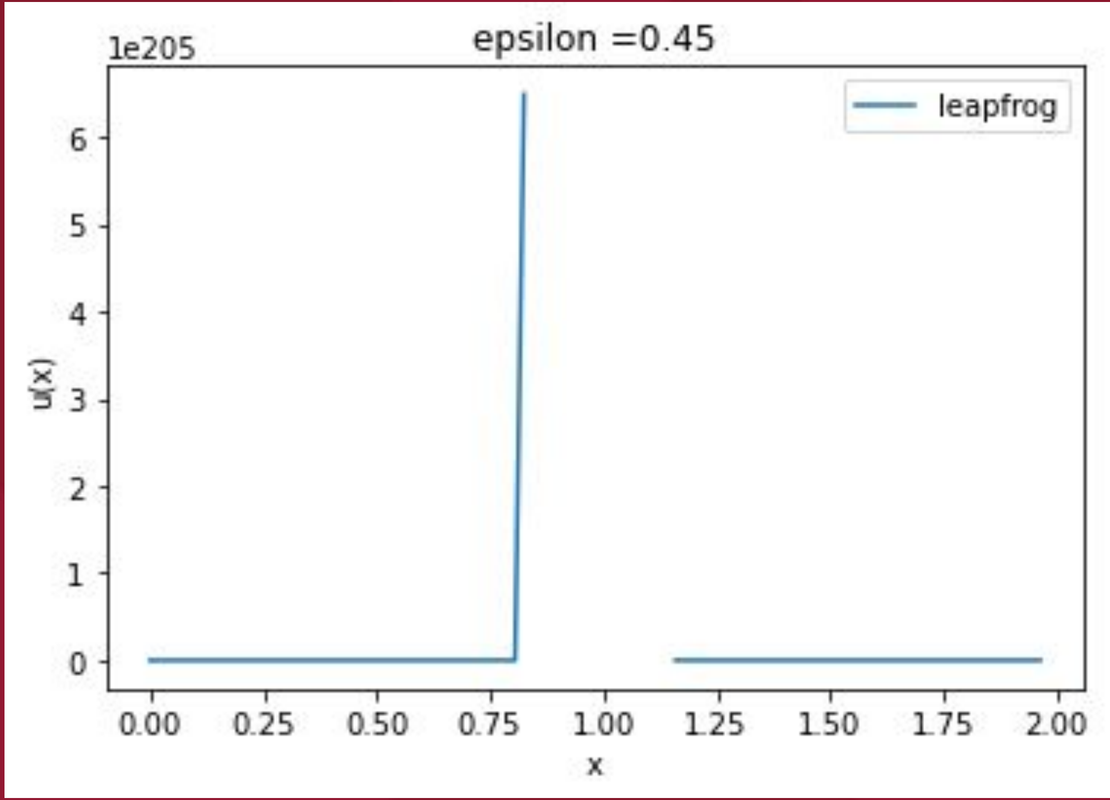
$$\boxed{u_i^{n+1} = u_i^{n-1} - \varepsilon u \frac{\Delta t}{\Delta x}(u_{i+1}^n - u_{i-1}^n)}$$

→ Ne nécessite que 2 solutions → faible consommation de mémoire

→ Condition de Courant → stable si $\dfrac{\Delta t}{\Delta x} < 1$

Résultats pour *leapfrog*




epsilon =0.45
1e205
leapfrog
u(x)
x
0
1
2
3
4
5
6
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

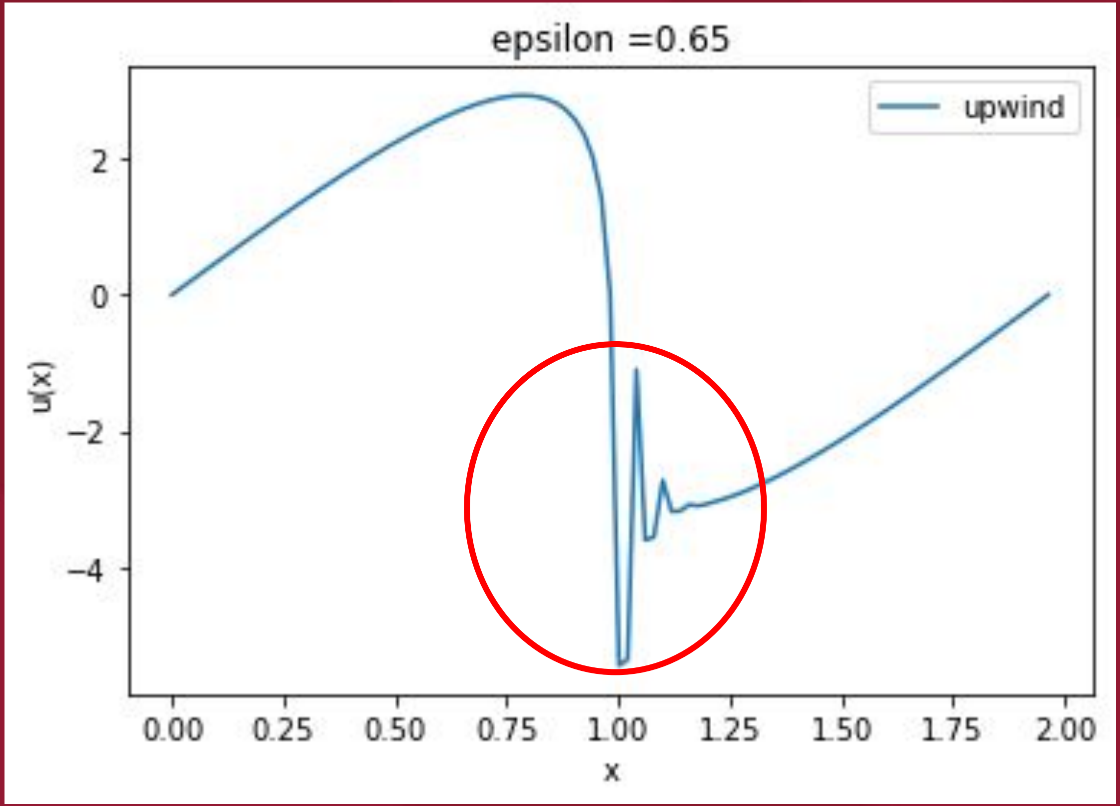
Méthode nº2: *up-wind*

→ 2 discrétisations possibles en fonction du signe de l'amplitude

$$\begin{cases} \dfrac{u_i^{n+1} - u_i^n}{\Delta t} + a\dfrac{u_i^n - u_{i-1}^n}{\Delta x} = 0 \quad , a > 0 \\[2ex] \dfrac{u_i^{n+1} - u_i^n}{\Delta t} + a\dfrac{u_{i+1}^n - u_i^n}{\Delta x} = 0 \quad , a < 0 \end{cases}$$

→ instabilités et oscillations

Résultats pour *up-wind*



Méthode nº3: *Lax-Wendroff*

- Méthode d'ordre 2 → plus grande précision

$$u_i^{n+1} = u_i^n - \frac{1}{2}\varepsilon u \frac{\Delta t}{\Delta x}(u_{i+1}^n - u_{i-1}^n) - \frac{1}{2}\left(\varepsilon u \frac{\Delta t}{\Delta x}\right)^2 (u_{i+1}^n + u_{i-1}^n - 2u_i^n)$$

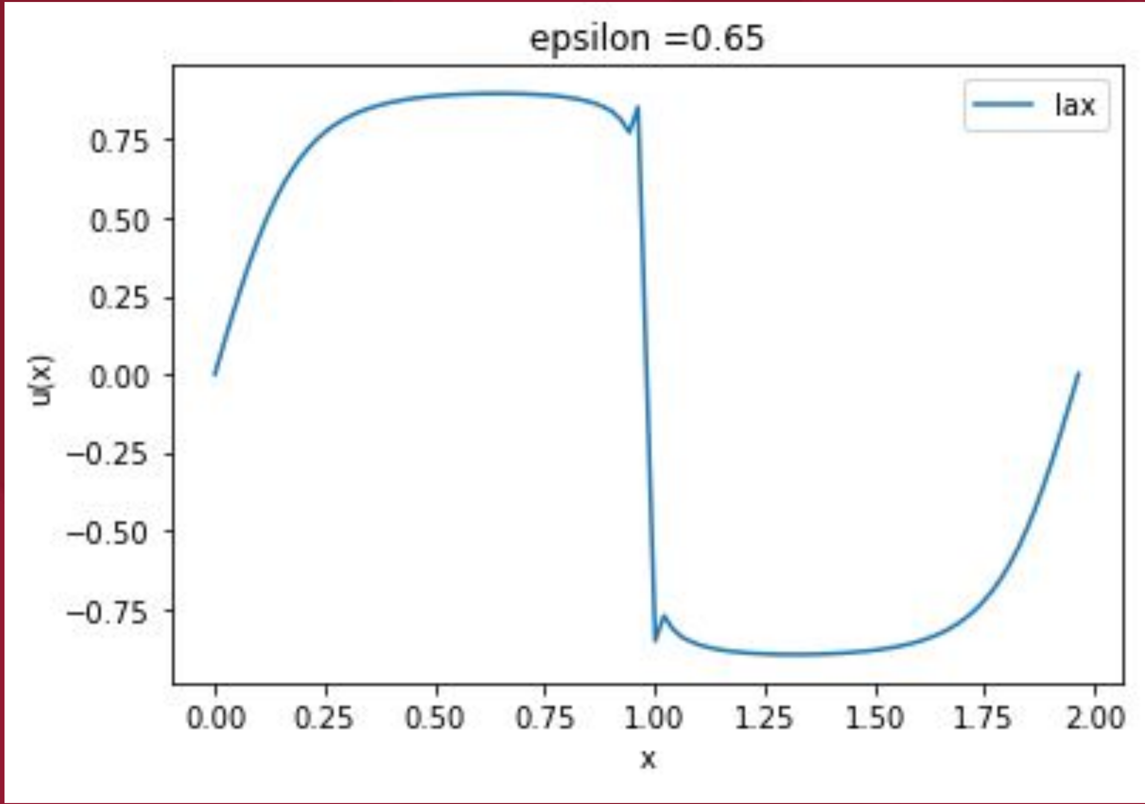
- Différences finies centrées

  → meilleure description du comportement autour du front de choc

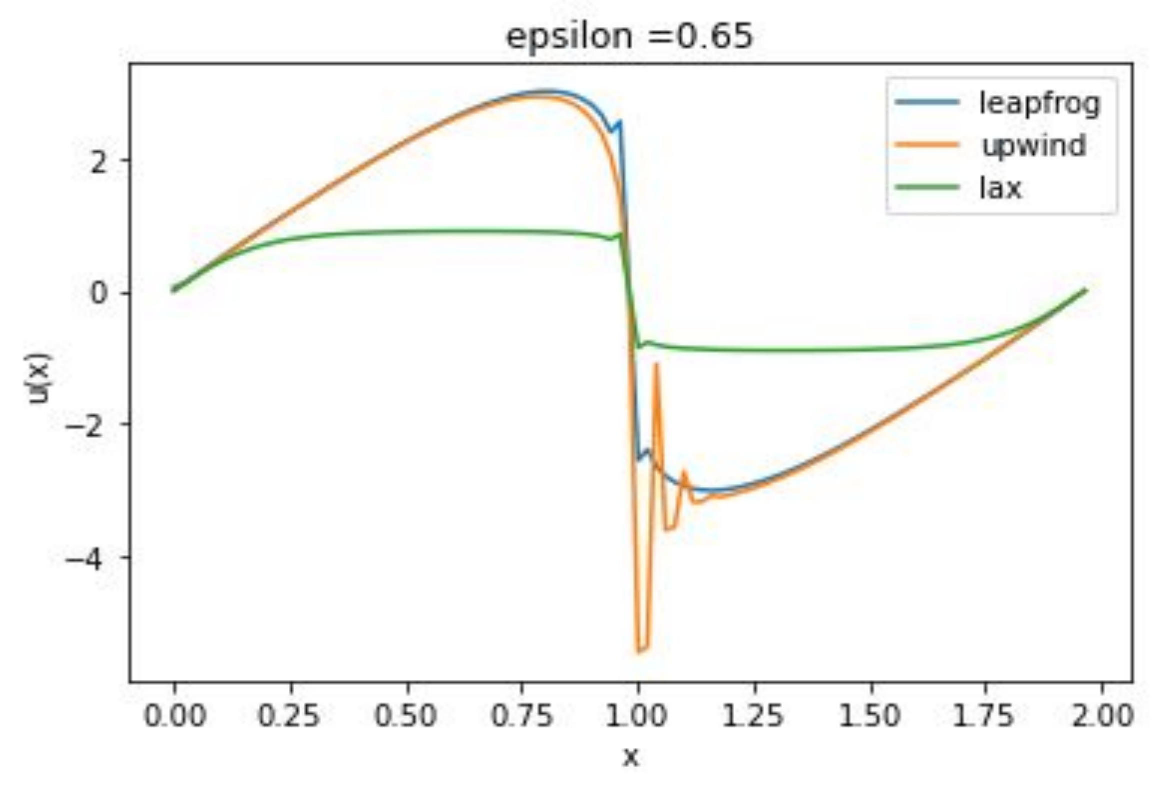
  → pas d'oscillations

- Coût informatique plus élevé

Résultats pour *Lax-Wendroff*



Comparaison des résultats



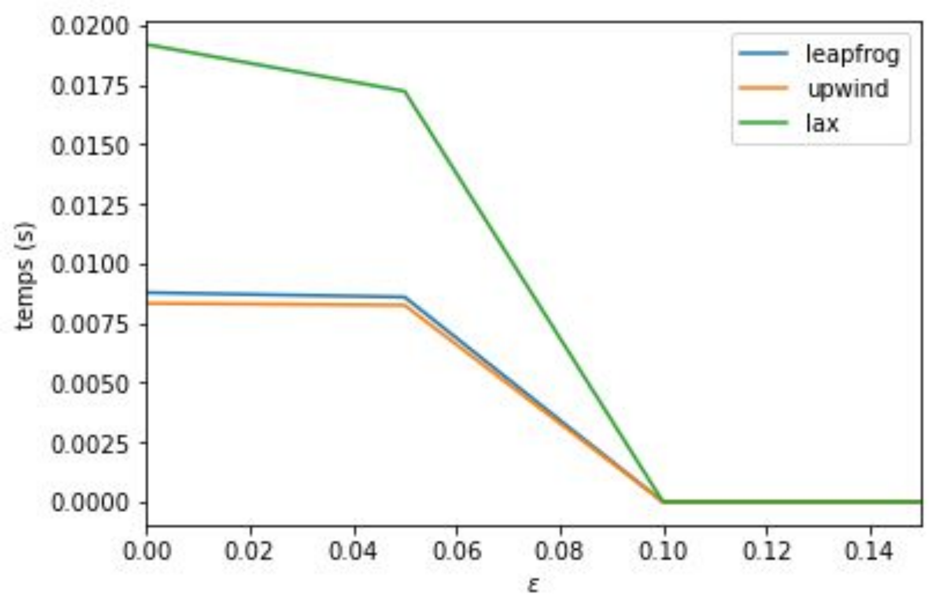
Conclusion

→ Lax-Wendroff

→ Plus précise

→ Plus stable

→ Plus coûteuse…



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Bibliographie

Le Veque Randall J. Numerical Methods for Conservation Laws, page 125. Birkhauser, 1992.

Discrétisation de *leapfrog*

$$\begin{cases} u(x+\Delta x,t) = u(x,t) + \dfrac{\partial u}{\partial x}\Delta x + \dots \\ \\ u(x-\Delta x,t) = u(x,t) - \dfrac{\partial u}{\partial x}\Delta x + \dots \end{cases}$$

$$\iff u(x+\Delta x,t) - u(x-\Delta x,t) = 2\frac{\partial u}{\partial x}\Delta x$$

$$\iff \frac{\partial u}{\partial x} = \frac{u(x+\Delta x,t) - u(x-\Delta x,t)}{2\Delta x}$$

$$\implies \frac{\partial u_i^n}{\partial t} = \frac{u_i^{n+1} - u_i^{n-1}}{2\Delta t}$$

$$\frac{u_i^{n+1} - u_i^{n-1}}{2\Delta t} + \varepsilon u \frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} = 0$$

$$\iff \boxed{u_i^{n+1} = u_i^{n-1} - \varepsilon u \frac{\Delta t}{\Delta x}(u_{i+1}^n - u_{i-1}^n)}$$

Discrétisation de *up-wind*

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + \varepsilon u \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$$

$$\iff \boxed{u_i^{n+1} = \left(1 - \varepsilon u \frac{\Delta t}{\Delta x}\right) u_i^n + \varepsilon u \frac{\Delta t}{\Delta x} u_{i-1}^n}$$

Discrétisation de *Lax-Wendroff*

$$u(x,t+\Delta t) = u(x,t) + \Delta t \frac{\partial u}{\partial t} + \frac{1}{2}\Delta t^2 \frac{\partial^2 u}{\partial t^2}$$

$$\rightarrow \frac{\partial u}{\partial t} = -\epsilon u \frac{\partial u}{\partial x} \iff \frac{\partial^2 u}{\partial t^2} = -\epsilon u \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x}\right) \iff \frac{\partial^2 u}{\partial t^2} = (\epsilon u)^2 \frac{\partial^2 u}{\partial x^2}$$

$$\implies u(x,t+\Delta t) = u(x,t) - \epsilon u \Delta t \frac{\partial u}{\partial x} + \frac{1}{2}(\epsilon u)^2 \Delta t^2 \frac{\partial^2 u}{\partial x^2}$$

$$\implies \begin{cases} \dfrac{\partial u}{\partial x} = \dfrac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} \\[2ex] \dfrac{\partial^2 u}{\partial x^2} = \dfrac{u_{i+1}^n + u_{i-1}^n - 2u_i^n}{\Delta x^2} \end{cases} \quad \boxed{\begin{aligned} \iff u_i^{n+1} = u_i^n &- \frac{1}{2}\epsilon u \frac{\Delta t}{\Delta x}(u_{i+1}^n - u_{i-1}^n) \\ &+ \frac{1}{2}(\epsilon u)^2 \frac{\Delta t^2}{\Delta x^2}(u_{i+1}^n + u_{i-1}^n - 2u_i^n) \end{aligned}}$$